

ASCII MASTER FLOW – PANEL 1/12: Siberian climate world distribution

SIBERIA -> largest single taiga band, continental interior

CANADA -> boreal belt from Rockies to Atlantic coast

SCANDINAVIA -> Sweden and Finland share the European taiga

SOUTHERN HEMISPHERE -> absent: no land broad enough at these latitudes

MUST REMEMBER: Cool temperate continental Siberian/D climate develops in high-latitude interiors with long severe winters, short summers, strong annual range, low precipitation and taiga conifers adapted to frozen soils and brief growth seasons.

ASCII MASTER FLOW – PANEL 2/12: Climate character profile

WINTER -> bitterly cold, long, sub-Arctic; very large annual range

SUMMER -> short, cool; brief warm growing season

RAINFALL -> low to moderate; light summer maximum

PRECIPITATION TYPE -> mostly snow in winter; frontal and convectional in summer

ASCII MASTER FLOW – PANEL 3/12: Conifer adaptation chain

CONICAL SHAPE -> sheds snow load, prevents branch breakage

NEEDLE LEAVES -> reduced surface area minimises transpiration loss

SHALLOW ROOTS -> exploit thin active layer above permafrost

RESULT -> conifers dominate where broadleaf trees cannot survive

ASCII MASTER FLOW – PANEL 4/12: Softwood economic chain

UNIFORM LOW-DIVERSITY STANDS -> mechanised extraction feasible

LONG-FIBRE SOFTWOOD -> ideal for pulp and paper manufacture

FROZEN GROUND AND RIVERS -> natural winter haulage surfaces

SPRING THAW -> river floating of logs to downstream mills

ASCII MASTER FLOW – PANEL 5/12: Carbon storage mechanism

COLD SLOWS DECOMPOSITION -> organic matter accumulates in soil and peat

PERMAFROST -> locks carbon in frozen ground for millennia

BELOW-GROUND CARBON -> disproportionately large compared to above-ground

RESULT -> boreal zone is a globally significant carbon reservoir

ASCII MASTER FLOW – PANEL 6/12: Fire-thaw feedback loop

WARMING -> lengthens fire season and thaws permafrost

FIRE -> releases stored carbon directly through combustion

THAW -> exposes frozen organic matter to decomposition

FEEDBACK -> released carbon causes further warming (self-reinforcing)

ASCII MASTER FLOW – PANEL 7/12: Permafrost engineering impact

THAWING PERMAFROST -> ground loses bearing capacity

ROADS AND RAILWAYS -> subsidence, cracking, realignment needed

PIPELINES -> stress fractures and leak risk

BUILDINGS -> foundation failure across settled sub-Arctic

ASCII MASTER FLOW – PANEL 8/12: Himalayan altitudinal zonation

SUBTROPICAL -> chir pine belt, 100-200 cm rain, fire-prone

TEMPERATE -> deodar, spruce in western Himalaya

SUBALPINE -> silver-fir (*Abies*) at 2200-3000 m, NW and NE

ALPINE -> bughyals/mergs: treeless pastures above treeline

ASCII MASTER FLOW – PANEL 9/12: Himalayan conifer identification

SILVER-FIR (*Abies*) -> 2200-3000 m; planking, pulp, paper, matches

SPRUCE (*Picea smithiana*) -> high wet-temperate belt; softwood

DEODAR (*Cedrus deodara*) -> western Himalaya; durable timber

CHIR PINE -> subtropical montane; resinous, fire-prone

ASCII MASTER FLOW – PANEL 10/12: Treeline and alpine pastures

CONIFEROUS FOREST -> gives way at the treeline altitude

ALPINE SCRUB -> dwarf shrubs in transition zone

BUGYALS/MERGS -> open grassy meadows for summer grazing

SNOWLINE -> permanent snow above the pastoral belt

CLOSE DISTINCTION: Taiga is boreal conifer forest south of tundra, permafrost extent varies and is not synonymous with all seasonally frozen ground, and continental subarctic climate differs from polar ET/EF thresholds.

ASCII MASTER FLOW – PANEL 11/12: Biome comparison strip

TAIGA -> high-latitude coniferous, below-ground carbon, softwood economy

STEPPE -> mid-latitude grassland, chernozem soil, mechanised grain

SAVANNA -> tropical grassland, laterised soil, pastoral subsistence

DISTINGUISHER -> latitude, rainfall regime, soil type and economic system

EVIDENCE LIMIT: Cold alone does not determine vegetation: moisture, active-layer depth, fire and disturbance matter; India's subalpine/alpine belts are elevation-controlled analogues with monsoonal and Himalayan differences.

ASCII MASTER FLOW – PANEL 12/12: Siberian climate answer spine

DEFINE -> Northern Hemisphere boreal coniferous belt with extreme continental range

LOCATE -> Siberia, Canada, Scandinavia; absent in Southern Hemisphere

EXPLAIN -> cold-adaptation, softwood economy, carbon storage, fire-thaw feedback

QUALIFY -> India's altitudinal analogue; transparent zero-direct-PYQ boundary